INCH-POUND

MIL-DTL-14645C
22 November 2002
SUPERSEDING
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29 January 1988

DETAIL SPECIFICATION

INSIGNIA, RANK AND GRADE, EMBROIDERED SHOULDER MARKS, AIR FORCE AND ARMY, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for Air Force and Army shoulder marks with embroidered insignia of rank and grade.

1.2 Classification. The shoulder marks are of the types, sizes, and classes noted below (see 6.2):

Type I - Air Force, Officer
- Size 1 - Large
- Size 2 - Small
- Class 1 - General
- Class 2 - Lieutenant General
- Class 3 - Major General
- Class 4 - Brigadier General
- Class 5 - Colonel
- Class 6 - Lieutenant Colonel
- Class 7 - Major
- Class 8 - Captain
- Class 9 - First Lieutenant
- Class 10 - Second Lieutenant

Type II - Army, Officer
- Size 1 - Large
- Size 2 - Small
- Class 1 - General
- Class 2 - Lieutenant General
- Class 3 - Major General
- Class 4 - Brigadier General

Comments, suggestions, or questions on this document should be addressed to Department of the Army, The Institute of Heraldry, Technical and Production Division, 9325 Gunston Road, Building 1466, Room S112, Fort Belvoir, VA 22060-5579 or haassw@tioh.belvoir.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

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Class 5 - Colonel
Class 6 - Lieutenant Colonel
Class 7 - Major
Class 8 - Captain
Class 9 - First Lieutenant
Class 10 - Second Lieutenant
Class 11 - Chief Warrant Officer W-4
Class 12 - Chief Warrant Officer W-3
Class 13 - Chief Warrant Officer W-2
Class 14 - Warrant Officer W-1
Class 15 - General of the Army
Class 16 - Chief Warrant Officer W-5

Type III - Army, Enlisted Personnel
Size 1 - Large
Size 2 - Small
Class 1 - Sergeant Major of the Army
Class 2 - Command Sergeant Major
Class 3 - Sergeant Major
Class 4 - First Sergeant
Class 5 - Master Sergeant
Class 6 - Sergeant First Class
Class 7 - Staff Sergeant
Class 8 - Sergeant
Class 9 - Corporal

Type IV - Air Force, Enlisted Personnel
Size 1 - Large
Size 2 - Small
Class 1 - Chief Master Sergeant of the Air Force
Class 2 - Chief Master Sergeant with First Sergeant Device
Class 3 - Chief Master Sergeant
Class 4 - Senior Master Sergeant with First Sergeant Device
Class 5 - Senior Master Sergeant
Class 6 - Master Sergeant with First Sergeant Device
Class 7 - Master Sergeant
Class 8 - Command Chief Master Sergeant

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless

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otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

V-T-285 - Thread, Polyester

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-823 - Cloth, Serge, Wool, Wool and Nylon, Polyester and Wool
MIL-E-17568 - Embroidery Materials, Metallic and Synthetic Metallic

FEDERAL STANDARDS

FED-STD-191 - Textile, Test Methods

(Copies of the above documents are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094 or http://astimage.daps.dla.mil/quicksearch/or www.dodssp.daps.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

THE INSTITUTE OF HERALDRY DRAWINGS AND CARTOONS (Details are italicized and enclosed in parentheses after the title of each drawing.)

B-3-86 - Insignia of Rank, Officer, Shoulder Mark, AF, Large *(General Officers, Colonel)*
B-3-86A - Insignia of Rank, Officer, Shoulder Mark, AF, Small *(General Officers, Colonel)*
B-3-87 - Insignia of Rank, Officer, Shoulder Mark, AF, Large *(Lieutenant Colonel, Major, Captain, First Lieutenant, Second Lieutenant)*
B-3-87A - Insignia of Rank, Officer, Shoulder Mark, AF, Small *(Lieutenant Colonel, Major, Captain, First Lieutenant, Second Lieutenant)*
B-3-92 - Insignia of Grade, Officer, Shoulder Marks, Army, General Officers, Large Size *(General of the Army, General, Lieutenant General, Major General, Brigadier General)*
B-3-93 - Insignia of Grade, Officer, Shoulder Marks, Army, Field and Company Grade, Large Size *(Colonel, Lieutenant Colonel, Major, Captain, First Lieutenant, Second Lieutenant)*
B-3-94 - Insignia of Grade, Officer, Shoulder Marks, Army, Warrant Officers, Large Size *(Master Warrant Officer MW-4/Chief Warrant Officer W-5, Chief Warrant Officer W-4, Chief Warrant Officer W-3, Chief Warrant Officer W-2, Warrant Officer W-1)*
B-3-95 - Insignia of Grade, Officer, Shoulder Marks, Army, General Officers, Small Size *(General of the Army, General, Lieutenant General, Major General, Brigadier General)*
B-3-96 - Insignia of Grade, Officer, Shoulder Marks, Army, Field and Company Grade,

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Small Size *(Colonel, Lieutenant Colonel, Major, Captain, First Lieutenant, Second Lieutenant)*
B-3-97 - Insignia of Grade, Officer, Shoulder Marks, Army, Warrant Officers, Small Size *(Master Warrant Officer MW-4/Chief Warrant Officer W-5, Chief Warrant Officer W-4, Chief Warrant Officer W-3, Chief Warrant Officer W-2, Warrant Officer W-1)*
B-3-98A - Insignia of Grade, Enlisted Personnel, Shoulder Marks, Sergeant Major of the Army, Large & Small Sizes
B-3-101 - Insignia of Grade, Enlisted Personnel, Shoulder Marks, Army, Large Size *(Command Sergeant Major, Sergeant Major, First Sergeant, Master Sergeant)*
B-3-101A - Insignia of Grade, Enlisted Personnel, Shoulder Marks, Army, Small Size *(Command Sergeant Major, Sergeant Major, First Sergeant, Master Sergeant)*
B-3-102 - Insignia of Grade, Enlisted Personnel, Shoulder Marks, Army, Large Size *(Sergeant First Class/Platoon Sergeant, Staff Sergeant, Sergeant, Corporal)*
B-3-102A - Insignia of Grade, Enlisted Personnel, Shoulder Marks, Army, Small Size *(Sergeant First Class/Platoon Sergeant, Staff Sergeant, Sergeant, Corporal)*
B-3-116B - Insignia of Grade, Shoulder Marks, Noncommissioned Officers, , AF, Large Size *(Chief Master Sergeant of the Air Force, Command Chief Master Sergeant, Chief Master Sergeant with First Sergeant Device, Chief Master Sergeant, Senior Master Sergeant with First Sergeant Device, Senior Master Sergeant, Master Sergeant with First Sergeant Device, Master Sergeant)*
B-3-116C - Insignia of Grade, Shoulder Marks, Noncommissioned Officers, AF, Small Size *(Chief Master Sergeant of the Air Force, Command Chief Master Sergeant, Chief Master Sergeant with First Sergeant Device, Chief Master Sergeant, Senior Master Sergeant with First Sergeant Device, Senior Master Sergeant, Master Sergeant with First Sergeant Device, Master Sergeant)*
B-3-117B - Insignia of Grade, Noncommissioned Officers, AF, Shoulder Marks, Large Size *(Construction & Details)*
B-3-117C - Insignia of Grade, Noncommissioned Officers, AF, Shoulder Marks, Small Size *(Construction & Details)*

(Copies of specifications, standards, drawings, and specification sheets required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of documents are those cited in the solicitation or contract.

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

AATCC-8 - Colorfastness to Crocking: AATCC Crockmeter Method
AATCC-96 - Dimensional Changes in Commercial Laundering of Woven and Knitted Fabrics Except Wool

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(These standards can be obtained from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709 or through the website at www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (formerly AMERICAN SOCIETY FOR QUALITY CONTROL)

ANSI/ASQC Z1.4 1993 - Sampling Procedures and Tables for Inspection by Attributes

(Requests for copies of ANSI/ASQC Z1.4 should be addressed to the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203-2914 or www.asq.org.

ASTM INTERNATIONAL (formerly AMERICAN SOCIETY FOR TESTING AND MATERIALS)

ASTM-D276 - Standard Test Methods for Identification of Fibers in Textiles
ASTM-D1907 - Standard Test Method for Linear Density of Yarn (Yarn Number) by the Skein Method
ASTM-D3776 - Standard Test Method for Mass per Unit Area (Weight) of Fabric
ASTM-D6193 - Standard Practice for Stitches and Seams

(These standards can be obtained from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or www.astm.org.)

THE COLOR ASSOCIATION OF THE UNITED STATES

United States Army Color Card, The Official Colors for Branches, Agencies, etc.

(Requests for copies of the above should be addressed to the Color Association of the United States, 315 West 39th Street, Suite 507, New York, New York 10018-1412 or www.colorassociation.com.

IRA GREEN, INC.

DOD Standard Shades for Heraldic Yarns for Embroidery and Ribbons, 1993 Edition

(Application for the above should be addressed to Ira Green, Inc., 177 Georgia Avenue, Providence, Rhode Island 02905.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.2 and 4.3.3.

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3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmental preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.3 Materials. Materials shall be as specified herein. Verification shall be in accordance with the standards and methods specified in 4.3.

3.3.1 Cloth, laminated. The laminated cloth shall be composed of one layer of shell (top) fabric and one layer of fusible fabric bonded together by heat and pressure. For Types I and IV insignia, the shell shall be polyester/wool cloth specified in 3.3.1.2, and the other layer shall be a fusible cloth specified in 3.3.1.1. For Types II and III insignia, the shell shall be a textured polyester specified in 3.3.1.3, and the other layer shall be a fusible cloth specified in 3.3.1.1. The two fabrics comprising the laminated cloth shall show good adhesion to one another. No adhesive from the fusible fabric shall show on the outside of the fusible cloth or the outside of the shell fabric. The shell fabric shall be free of bubbles or puckering before and after wet dry cleaning. The laminated cloth shall provide a firm base for embroidering yet be adequately pliable to permit hemming, stitching, and turning.

3.3.1.1 Cloth, fusible. The fusible cloth shall be a cotton or polyester/cotton cloth coated with a polyamide film or any adhesive that guarantees a permanent bond and meets the requirements stated above. The weave shall be plain and the weight shall be 4 to 5 oz. per square yard. The maximum shrinkage shall be 2% in the warp and filling. The color shall be white or greige.

3.3.1.2 Cloth, polyester/wool (shell fabric). The top layer of the laminated cloth for Types I and IV shoulder marks shall be a Type III, Class 4 polyester/wool serge having the following physical requirements.

Weight Minimum: 9.5 ounces [per square yard]
Yarn Ply:
  Warp: 2
  Filling: 2
Yarns Per Inch:
  Warp: 78
  Filling: 66

The color shall be a good match to Air Force Shade 1620 when viewed under natural (north sky) daylight or filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7000 ± 500 K, with illumination of 100 ± 20 foot candles. The color shall also be a good match to the specified color under incandescent lamplight at 2300 ± 100 K. The cloth shall show good fastness to wet-dry cleaning, perspiration, light, laundering, and crocking.

3.3.1.3 Cloth, textured polyester (shell fabric). The top layer of the laminated cloth for Types II and III shoulder marks shall be a textured polyester serge having the following physical requirements:

Weight Minimum: 6.5 ± 0.5 ounces [per square yard]
Yarn Ply:
  Warp: 2
  Filling: 2

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Yarns Per Inch:
Warp: 70
Filling: 66
Weave: 2 up and 2 down, 4 harness right twill

The color of the shell fabric shall be U.S. Army Black, Cable No. 65018 (see the U.S. Army Color Card) when viewed under natural (north sky) daylight or filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7000 ± 500 K, with illumination of 100 ± 20 foot candles. The color shall also be a good match to the specified color under incandescent lamplight at 2300 ± 100 K. The cloth shall show good fastness to wet-dry cleaning, perspiration, light, laundering, and crocking.

3.3.2 Thread. The thread for hemming the shoulder mark and sewing the seams shall be polyester and conform to Type I, Class 1, Letter Size AA, 2-ply of V-T-285. The thread shall match the color of the shell fabric.

3.3.3 Embroidery yarn.

3.3.3.1 Synthetic metallic embroidery yarn. The synthetic metallic yarn for embroidery shall conform to Type III, Class 8 or 9 of MIL-E-17568 with the following exceptions.

a. The finished yarn size shall be 260 denier ± 15 denier.

b. The yarn shall have sufficient turns per inch to ensure proper uniformity of the metallic ribbon wrapped around the core and can be used on any type of embroidery machine to meet specification requirements. The metallic yarn shall match the color specified on the applicable drawing or cartoon.

3.3.3.2 Rayon or polyester embroidery yarn. The embroidery yarn shall be a semi-dull continuous filament, regenerated cellulose type rayon, 2-ply Schiffli type or Multihead yarn. The use of 2-ply polyester is authorized in lieu of rayon. Denier of the yarn shall be as specified on the cartoon or drawing. Comparable sizes are as follows:

SCHIFFLI		**MULTIHEAD**	
Rayon	Polyester	Rayon	Polyester
150/2	135/2	150/2 or 35 weight	-----
100/2	90/2	120/2 or 40 weight	120/2 or 40 weight
75/2	70/2	-----	95/2

Yarn color shall be:

Graphite Blue HCF 67181
White HA 67101
Goldenlight HG 67107
or Black HAN 67138

as specified. The yarn shall show good colorfastness to wet dry-cleaning and light.

3.4 Design. The design of each shoulder mark shall be as shown on the applicable drawing.

3.5 Construction.

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3.5.1 Embroidery. The device(s) and stripe(s) for each class of shoulder mark shall be Schiffli or Multihead embroidered on the face side of the shell fabric. The embroidery stitches shall be of the type, color (see DOD Standard Shades for Heraldic Yarns for Embroidery and Ribbons), length, minimum number, and density specified by the applicable cartoon. Yarn specified in 3.3.3.1 and 3.3.3.2 shall be used as shown in TABLE I.

TABLE I. **Use of embroidery yarns.**

TYPE, CLASS	YARN(S) USED
Type I:	
Classes 1 thru 6, 8, and 9	all silver metallic **1/**
Classes 7 & 10	silver metallic **1/** for stripe; gold metallic **2/** for device
Type II:	
Classes 1 thru 4	white rayon or polyester for stars; gold metallic **2/** for stripe
Classes 7 & 10	all gold metallic **2/**
Classes 5, 6, 8, & 9	silver metallic **1/** for device; gold metallic **2/** for stripe
Classes 11 thru 14, 16	silver metallic **1/** & black rayon or polyester for device; gold metallic **2/** for stripe
Class 15	white rayon or polyester for stars; gold metallic **2/** for eagle device & stripe
Type III:	all gold metallic **2/**
Type IV	silver metallic **1/** for device; graphite blue rayon or polyester for border stitch

1/ Silver metallic embroidery shall have an underlay of white rayon or polyester
2/ Gold metallic embroidery shall have an underlay of goldenlight rayon or polyester

Sufficient base cloth shall be allowed in all directions so shoulder marks can be properly hemmed and joined together.

3.5.2 Cutting. After insignia have been embroidered, material shall be cut so that the finished shoulder mark will conform to the measurements specified on the applicable drawing.

3.5.3 Hemming. The hemming shall be executed at the location indicated on the applicable drawing or cartoon. Each hem shall have a minimum of 10 and a maximum of 12 stitches per inch using a type 301 stitch of ASTM-D6193. All ends of stitching not caught in the seam shall be backstitched a minimum of 1/4 inch. Hems shall be 3/8 inch + 1/8 - 0 inch wide, and stitching shall be located 1/8 inch + 0 - 1/32 inch from the turned edge.

3.5.4 Seam stitching. Seam stitching shall be in accordance with Type 301 of ASTM D-6193. All ends of stitching shall be backstitched not less than 1/4 inch. Thread breaks shall be backstitched not less than 1/2 inch at each break. Thread tension shall be maintained equally for top and bottom thread. Seams shall be sewn with not less than 10 and not more than 12 stitches per inch. All thread ends shall be trimmed.

3.5.5 Pressing. The finished shoulder mark shall be turned and pressed so that the item has the shape shown on the applicable drawing.

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3.6 Tolerances. Unless otherwise specified herein or on the applicable drawing, dimensions of the finished shoulder mark and its visible components shall be within a tolerance of ± 1/32 inch (see 4.3.3).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. Inspection and testing of the first article (see 3.1) shall be made of a completely finished item or items for all provisions of this specification applicable to the end product examination and tests (see 4.3.2 and 4.3.3).

b. Conformance inspection. Inspection shall be in accordance with the provisions set forth herein, except where otherwise indicated (see 6.2). Sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4.

4.2 Inspection conditions. Unless otherwise stated, all inspections shall be performed in accordance with the test conditions specified in 4.3.

4.3 Methods of inspection.

4.3.1 Testing of components. Except as otherwise stated in TABLE II, testing of components shall be in accordance with the stated test method from FED-STD-191.

TABLE II. **Testing of components.**

COMPONENT	CHARACTERISTIC	RQM'T¶	TEST METHOD	RESULTS REPORTED AS
Laminated cloth	bubbles/puckering before & after wet dry cleaning	3.3.1	visual	pass or fail
Fusible cloth	amount of adhesive	3.3.1.1	visual	pass or fail
	material identification	3.3.1.1	ASTM-D276, **1/**	pass or fail
	weave	3.3.1.1	visual	pass or fail
	weight	3.3.1.1	ASTM-D3776, **1/**	nearest 0.1 oz/sq yd
	shrinkage	3.3.1.1	AATCC-96, Vc, **1/**	nearest 0.1 % in each direction
Polyester/wool shell fabric	material identification	3.3.1.2	ASTM-D276, **1/**	pass or fail
	colorfastness to:			
	wet dry-cleaning	3.3.1.2	5622, **1/**	pass or fail
	light	3.3.1.2	5662, **1/**	pass or fail
	crocking	3.3.1.2	AATCC-8, **1/**	pass or fail
Textured poly-ester shell fabric	material identification	3.3.1.3	ASTM-D276, **1/**	pass or fail
	colorfastness to:			
	wet dry-cleaning	3.3.1.3	5622, **1/**	pass or fail
	light	3.3.1.3	5662, **1/**	pass or fail

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TABLE II. **Testing of components.** - continued

COMPONENT	CHARACTERISTIC	RQM'T	TEST METHOD	RESULTS REPORTED AS
	crocking	3.3.1.3	AATCC-8, **1/**	pass or fail
Synthetic metallic embroidery yarn	material identification	3.3.3.1	MIL-E-17568, **1/**	pass or fail
	denier	3.3.3.1	ASTM-D1907, **1/**	pass or fail
	uniformity of ribbon around core	3.3.3.1	visual	pass or fail
Rayon or polyester embroidery yarn	material identification	3.3.3.2	ASTM-D276; **1/**	pass or fail
	ply	3.3.3.2	visual	pass or fail
	denier	3.3.3.2	ASTM-D1907, **1/**	pass or fail
	presence of sulfur	3.3.3.2	**1/**	pass or fail
	colorfastness to:			
	wet dry-cleaning	3.3.3.2	5622, **1/**	pass or fail
	light	3.3.3.2	5662, **1/**	pass or fail

1/ Unless otherwise specified, a contractor's certification of compliance may be accepted as evidence of conformance.

4 3.2 Visual examination of the end item. Defects found during this examination shall be classified according to Table III. The sample unit for this examination shall be one finished pair of shoulder marks. Normal inspection distance for workmanship shall be 18 to 24 inches with illumination equal to average daylight and arranged so as to avoid as much reflected light as possible. Defects designated by an asterisk (✳) shall be classified as major when seriously affecting appearance or serviceability and minor when affecting appearance or serviceability but not seriously.

TABLE III **Visual examination of the end item.**

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	(✳)	MINOR
Laminated cloth	Bubbles and/or puckering visible on shell fabric before or after dry cleaning.	X		
Fusible cloth	Material not as specified			
Shell fabrics	Material not as specified.	X		
	Manufacturing defect(s) such as hole, float, broken end or pick, slub, lump, knot, streak			
Design	Any shift affecting appearance.	X		
	Not in accordance with cartoon or drawing. . .	X		
Embroidered device and stripe	Missing, incomplete, wrongly positioned, or distorted.	X		
	Stripe does not extend beyond creased line when applicable	X		

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TABLE III **Visual examination of the end item** - continued.

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	(*)	MINOR
	Not underlaid to give a well raised effect. . . .	X		
	Incorrect embroidery yarn.			
	Incorrect color of yarn.	X		
	Too loose tension gives a loopy appearance to stitches.			
	Too tight tension pulls stitches, puckers cloth, and may distort device.			
	Ragged, broken, or loose stitches.			
Thread	Not a good color match to shell fabric.			
Hems and seams	Loose ends not backstitched a minimum of 1/4 inch.			
	Thread breaks not backstitched a minimum of 1/2 inch.			
	Tension too loose or too tight.			
	Less than 10 or more than 12 stitches per inch.			
	Any open seam; any unstitched or unattached area of hem.	X		
	Seam stitching crooked or irregular.			
	Hem stitching crooked or irregular.			
	Finished hem not 3/8 to 1/2 inch in width. . . .			
	Hem stitching not 3/32 to 1/8 inch from turned edge.	X		
	Incorrect stitch type in seam.			
	Incorrect stitch type in hem.			
Pressing	Singed or scorched.	X		
	Inappropriate creases or gloss marks clearly visible.	X		
	Misshapen by pressing.			
	Color change clearly noticeable.	X		
	Not pressed flat.			
Workmanship	Spotted, stained, or soiled.	X		
	Not packaged in pairs when required by contract or solicitation	X		
	Types I & II, Class 5 not packaged in pairs of left and right.	X		
	Shoulder marks forming a pair do not match, e.g., cloth color; overall dimensions; shape, size, color, placement, and appearance of device(s) and/or stripe(s).	X		
	Basting stitches or cutting marks show on outside of shoulder mark.			

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4.3.3 Examination for dimensions. Measurements shall be taken of all points, lines, areas, curves, radii, and spaces for which dimensions are specified in the drawing. Tolerances shall be as specified in 3.6. Any dimension that is not within the specified tolerance shall be classified as a major defect.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The embroidered shoulder marks covered by this specification are intended to be worn by personnel of the Army and Air Force to indicate rank or grade.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, size, and class required (see 1.2).
c. Issue of DoDiss to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1).
d. Drawing/cartoon number (see 2.2.2).
e. When a first article is not required (see 3.1).
f. Exceptions to or deviations, if any, from inspection requirements (see 4.1).
g. Packaging requirements (see 5.1).

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first production sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.5 Government furnished property. The contracting officer should arrange to furnish the cartoons and drawings required in 3.3, 3.4, and 3.5.

6.6 Suggested packaging. Packaging, packing, preservation, marking, and any acceptance criteria will be in accordance with 5.1. Note that acceptable packaging may take a number of different forms commercial wrapping paper; clear blister or "bubble" wrap; shrink wrap; bags made of commercial grade paper, clear plastic, or clear blister/"bubble" pack closed by tying, taping, or heat-sealing, as appropriate. The intention is to keep a stated number of units of issue together for counting, marking, and subsequent packaging or packing and to do so in a manner that does no damage to the items during handling, shipping, and storing. Suggested packaging is given below:

One pair of shoulder marks of one type, size, and class only may be placed in a snug-fitting bag. Note that for Types I and II, Class 5, a pair consists of one left and one right shoulder mark.

6.7 Subject term (key word) listing.

Cartoon
Colorfastness
Denier
Fusible
Laminated
Multihead
Polyamide
Schiffli
Sulfur
Textured polyester

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - IH
Air Force - 11

Preparing activity:
Army - IH

Review activities:
DLA - CT

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-14645C	2. DOCUMENT DATE (YYYYMMDD) 20021122

3. DOCUMENT TITLE INSIGNIA, RANK AND GRADE, EMBROIDERED SHOULDER MARKS, AIR FORCE AND ARMY, GENERAL SPECIFICATION FOR

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
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8. PREPARING ACTIVITY

NAME The Institute of Heraldry, Dept. of the Army	b. TELEPHONE *Include Area Code)* (1) Commercial 703-806-4982 (2) AUTOVON 656-4982
c. ADDRESS *(Include Zip Code)* Technical and Production Division 9325 Gunston Rd., Bldg. 1466, Rm S-112 Fort Belvoir, Virginia 22060-5579	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman road, Suite 2533, Ft. Belvoir, VA 22060-2533 Telephone (703) 767-6888 AUTOVON 427-6888

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